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Autonomous Predictive Tracking and Laser Neutralization System for Flies

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1 Abstract

Flies constitute a persistent problem in domestic, industrial, and laboratory environments due to their role in contamination, disease transmission, and disruption of controlled spaces. Unlike many other insects, flies exhibit extremely rapid, irregular, and highly unpredictable flight behavior. These characteristics make them particularly difficult to track and respond to using existing automated systems, which typically assume smoother and more predictable motion patterns. current tracking and interception technologies, while effective for slower or more structured targets, fail to operate reliably in the presence of fly dynamics due to latency, limited prediction accuracy, and insufficient adaptability to abrupt motion changes. Consequently, there is a need for autonomous systems specifically designed to address the unique behavioral properties of flies. This project proposes an autonomous predictive tracking framework tailored to fly motion. by combining real-time visual sensing with a Transformer-based temporal prediction model, the system aims to learn complex motion patterns and anticipate future positions despite apparent randomness. The work focuses on the conceptual design, underlying theory, and safety considerations of such a system, addressing a gap not adequately covered by existing solutions. Our project is available in our GitHub Repository.

2 Introduction

In everyday human environments, the presence of flies remains a persistent and widely overlooked problem. Flies are commonly found in homes, food-handling areas, laboratories, and medical facilities, where they contribute to contamination, disease transmission, and disruption of controlled spaces. Despite their small size, flies can have a significant impact on hygiene, safety, and comfort, making effective mitigation an important practical concern.

Through observation of such environments, it becomes evident that existing fly-control approaches are often inadequate. Manual intervention is inefficient and unreliable, while chemical solutions raise concerns related to health, environmental impact, and long-term effectiveness. These limitations highlight the need for alternative solutions that can operate autonomously and continuously without human involvement.

From an engineering perspective, flies pose a particularly challenging target for automated systems. Unlike many other insects or moving objects, flies exhibit extremely fast, irregular, and highly unpredictable flight behavior. Their motion includes sudden directional changes and short reaction times, which render conventional tracking and interception methods ineffective. Systems that rely on reactive control or short-term prediction struggle to respond accurately within the required time constraints. Motivated by these challenges, this project proposes the development of an autonomous system designed specifically to address fly dynamics. Rather than aiming at the current observed position of the fly, the system focuses on predicting its future location and directing a precise response toward that predicted position. By combining real-time visual sensing with temporal motion modeling, the system seeks to overcome latency and improve targeting reliability in highly dynamic scenarios.

To enable accurate prediction of complex and non-linear motion patterns, the proposed framework employs a Transformer-based sequence model trained on observed fly trajectories. The predicted future position serves as the basis for directing a localized neutralization mechanism, allowing effective interaction despite the fly's rapid and unpredictable movement. Emphasis is placed on conceptual system design, robustness, and safety considerations.

In the subsequent chapters of this book, we explore the theoretical and practical aspects of autonomous predictive tracking for flies. First, we review related work in object tracking and insect monitoring. We then present the background concepts required to understand fly motion characteristics and predictive modeling approaches. Building on this foundation, we describe the proposed system architecture and data flow, followed by a discussion of evaluation considerations and future improvements.

Figure 1: A human attempting to catch a fast-moving housefly in a kitchen environment. The image illustrates the difficulty of intercepting highly agile and unpredictable insects, motivating the need for autonomous predictive tracking systems.

Figure 2: A fly captured by a Venus flytrap (*Dionaea muscipula*). In nature, successful interception relies on rapid sensing and precise timing—capabilities that inspire engineered systems for autonomous fly detection, tracking, and prediction.

3 Background and Related work

This chapter presents the background concepts and related research relevant to the development of an autonomous predictive tracking system for flies. It focuses on prior work in object detection, motion tracking, and prediction, with emphasis on methods applicable to fast and irregular targets. The reviewed material highlights the limitations of existing approaches and provides the foundation for the predictive framework proposed in this project.

3.1 Safety and Ethical Considerations

Autonomous systems that interact physically with their environment must incorporate safety and ethical constraints as core design requirements. This is particularly important for systems that rely on visual perception and autonomous decision-making, where uncertainty and unexpected conditions may lead to unintended behavior. The following subsections outline the primary risks associated with the proposed system and the considerations required to mitigate them.

3.1.1 Risk of Human Presence in the Field of View

One of the most critical risks arises from the possible presence of people within the camera's field of view. Since the system relies on visual input from a monitored environment, humans may unintentionally appear in the scene, either partially or fully. Any autonomous response in such scenarios could pose a significant safety hazard.

To address this risk, the system must be designed to identify human presence and prevent any activation when people are detected within the operational area. This requirement ensures that the system operates exclusively in controlled environments and prioritizes human safety above all other objectives.

3.1.2 Perception and Prediction Errors

Autonomous systems are inherently subject to uncertainty due to sensor noise, lighting variations, occlusions, and limitations of detection and prediction models. Errors in perception or motion prediction may result in incorrect estimation of target position or behavior.

Such errors introduce the risk of unintended responses. To mitigate this, system operation should be conditioned on confidence measures and validation mechanisms, ensuring that actions are only considered when prediction reliability is sufficiently high. In cases of uncertainty, the system should default to a non-action state.

3.1.3 Operation in Uncontrolled Environments

Another potential risk is the use of the system outside of its intended operational context. Environments that are public, dynamic, or insufficiently constrained may violate the assumptions under which the system is designed to function safely.

Ethical deployment therefore requires strict limitations on where and how the system may be used. The system should be restricted to predefined, controlled environments, and its functionality should not extend to open or populated spaces.

3.1.4 System Malfunction and Fail-Safe Behavior

Hardware failures, software bugs, or unexpected runtime conditions may lead to system malfunction. Without appropriate safeguards, such failures could result in unpredictable behavior. To address this risk, the system design must incorporate fail-safe mechanisms that ensure safe shutdown or inactivity in the event of malfunction. Conservative system behavior in failure scenarios is essential to maintaining safety and reliability.

3.1.5 Ethical Responsibility and Accountability

Beyond technical risks, ethical responsibility plays a central role in the design of autonomous systems. Clear documentation of system behavior, operational assumptions, and limitations is necessary to ensure transparency and accountability.

Designers and operators must remain aware of the ethical implications of autonomous decision-making and ensure that the system's capabilities are not misused. Emphasizing accountability and responsible use helps align technological development with societal expectations and ethical standards.

3.2 Fly Motion Characteristics

Flies exhibit flight behavior that poses a significant challenge for autonomous tracking systems. Their motion is characterized by high speed, rapid acceleration, and abrupt changes in direction over very short time intervals. Unlike many other insects or moving targets, flies do not follow smooth or easily predictable trajectories.

At short time scales, fly motion often appears irregular and non-linear, making it difficult to approximate using classical kinematic models. Sudden turns and direction reversals reduce the effectiveness of tracking methods that assume gradual changes in velocity or direction. These characteristics significantly increase the difficulty of maintaining accurate localization across consecutive frames.

In addition to their dynamic motion, flies are small in size relative to the camera field of view. As a result, they occupy only a limited number of pixels, which makes detection and tracking sensitive to noise, motion blur, and lighting variations. When combined with fast movement, these factors further degrade tracking reliability.

Beyond these intrinsic motion properties, fly behavior is also strongly influenced by environmental and biological factors. Studies have shown that temperature, humidity, light intensity, and air flow affect flight activity, speed, and turning frequency. For instance, higher temperatures generally increase activity levels, while variations in illumination and humidity modify exploratory and evasive behaviors. Furthermore, different fly species exhibit distinct flight statistics due to differences in body mass, wingbeat frequency, and sensory response mechanisms. Even within a single species, individual variability contributes to non-uniform motion patterns.

These dependencies introduce non-stationarity into the motion dynamics, making fly trajectories highly context-dependent and inherently stochastic. As a result, reactive targeting strategies that rely solely on the current observed position of the fly are fundamentally limited. By the time a response is executed, the fly may have already moved to a new location. This motivates the need for predictive approaches that estimate future positions and form the basis for the system design presented in later chapters [4, 8, 13].

3.3 Vision-Based Object Detection

Vision-based object detection is a fundamental component of autonomous systems that interact with dynamic environments. Using camera input, detection algorithms aim to localize objects of interest within individual frames, typically by identifying bounding regions or key points. In recent years, advances in image processing and machine learning have significantly improved detection accuracy and robustness.
Detecting flies presents a particular challenge due to their small size, fast movement, and frequent overlap with complex backgrounds. Unlike large or slow-moving objects, flies often occupy only a small number of pixels and are easily affected by motion blur and lighting variations. These factors necessitate detection approaches that are both sensitive and computationally efficient.

3.3.1 Small Object Detection

Small object detection refers to the problem of identifying targets that occupy a limited area in the image frame. Such targets are more susceptible to noise, resolution limitations, and background interference. Standard detection algorithms often struggle in these conditions, as distinguishing features may be weak or inconsistent across frames. For flies, small object detection is further complicated by rapid motion and frequent changes in appearance due to orientation and wing movement. These challenges reduce detection stability and increase the likelihood of missed or incorrect detections, which directly impacts downstream tracking and prediction performance [2].

3.3.2 Insect Detection in Cluttered Environments

In many real-world scenarios, house flies appear against cluttered and dynamic backgrounds, including textured surfaces, moving shadows, and variable illumination. Such background complexity can significantly reduce detection performance, particularly when the target exhibits low visual contrast. Robust detection in these conditions requires advanced approaches, such as neural-network-based models capable of distinguishing flies from background noise and adapting to changing lighting and environmental patterns.
Traditional insect detection methods often rely on background subtraction or motion cues; however, these approaches can become unreliable under variable environmental conditions. Achieving robust detection in cluttered environments remains a critical challenge and a key consideration in the design of autonomous fly tracking systems, as demonstrated in prior studies on high-resolution fly tracking and multi-camera 3D motion analysis [**?**, 12].

3.4 Human Detection from Visual Data

Human detection from images and video is a well-studied problem in computer vision and serves as a fundamental component in many vision-based systems. Research in this area has focused on identifying the presence of people within a scene despite variations in pose, scale, lighting conditions, and partial occlusions [3, 10].

Early approaches relied on hand-crafted visual features combined with classical classifiers, while more recent methods employ learning-based models that automatically extract discriminative features from image data. Modern detection algorithms are capable of robustly identifying humans across diverse environments and are commonly used as a supporting module in larger autonomous systems. These advances make human detection a reliable and practical component for integration into safety-aware system designs. In the proposed system, both fly detection and human detection rely on learning-based computer vision models. While fly detection provides the input required for tracking and motion prediction, human detection operates as an independent safety mechanism. The detection of human presence overrides all other system components and disables autonomous operation.

3.5 Object Tracking Methods

Object tracking aims to maintain a consistent estimate of a target's position over time by associating detections across consecutive frames. Tracking methods typically incorporate motion models to predict the target's next position and reduce uncertainty. These techniques are widely used in robotics, surveillance, and autonomous navigation.
While effective for many applications, classical tracking approaches are often designed under assumptions of smooth and continuous motion. When applied to flies, these assumptions are frequently violated, resulting in degraded tracking accuracy and frequent loss of the target.

3.5.1 Kalman Filter

The Kalman filter is a widely used state-estimation technique that assumes linear motion and Gaussian noise. It combines model-based predictions with noisy measurements to produce an optimal state estimate under these assumptions.
Due to the highly irregular and non-linear nature of fly motion, these assumptions are frequently violated, leading to inaccurate predictions and reduced tracking performance.

3.5.2 Particle Filter

Particle filters extend classical approaches by representing the target state using a set of weighted samples, allowing them to model non-linear and non-Gaussian motion. This makes them more flexible than Kalman filters in dynamic environments.
However, particle filters are computationally expensive and still rely on motion models that may not adequately capture the extreme variability of fly trajectories, limiting their practicality in real-time systems.

3.6 Limitations of Reactive Tracking Systems

Reactive tracking systems respond based solely on the current observed position of a target. While this approach can be effective for slow or predictable objects, it becomes inadequate when system latency and target dynamics are significant. This limitation motivates the use of prediction not as an enhancement, but as a necessity for compensating system latency inherent in sensing, processing, and actuation pipelines. For flies, the delay between sensing, processing, and response often exceeds the time required for the target to change position. As a result, actions aimed at the current position are frequently misaligned with the actual target location at execution time [1].

3.7 Motion Prediction Models

Motion prediction models aim to estimate a target's future position based on past observations. By anticipating future motion, predictive systems can compensate for latency and improve response accuracy.

3.7.1 Classical Motion Prediction

Classical prediction methods use simple mathematical formulations, such as linear or polynomial extrapolation, to forecast future positions. These approaches are computationally efficient but rely on assumptions of smooth and continuous motion.
Such assumptions are often violated in fly motion, resulting in poor prediction accuracy even over short time horizons.

3.7.2 Learning-Based Prediction Models

- **Recurrent Neural Networks** Recurrent neural networks, including LSTM and GRU architectures, model temporal dependencies by processing sequences of observations. While they can capture more complex dynamics than classical models, they may struggle with long-range dependencies and rapidly changing motion.

- **Transformer-Based Temporal Models** Transformer-based models use attention mechanisms to capture relationships across time steps, enabling efficient modeling of both short- and long-term dependencies. Recent research has demonstrated their effectiveness in trajectory prediction tasks under noisy and dynamic conditions. These properties make Transformer-based models particularly suitable for predicting the highly irregular motion of flies and motivate their use in the proposed system.

3.8 Autonomous Targeting and Predictive Control

Autonomous targeting systems integrate perception, prediction, and control to interact with dynamic targets. Predictive control strategies rely on future state estimates rather than current observations, improving robustness and accuracy.
In the context of fly tracking, predictive targeting enables the system to direct its response toward a predicted future position, increasing effectiveness despite rapid and unpredictable motion.

3.9 Related Work

Existing research relevant to this work can be categorized into four domains: high-speed insect flight tracking, learning-based visual trajectory prediction, laser-based insect monitoring and discrimination systems, and embedded vision platforms for real-time laser neutralization.

3.9.1 TrajLearn: Transformer-Based Trajectory Prediction

Recent advances in trajectory forecasting have demonstrated the power of transformer-based architectures for modeling complex motion patterns from historical sequences. For instance, Amir et al. (2025) introduced TrajLearn, a generative trajectory prediction framework that leverages a transformer encoder-decoder to capture long-range spatial-temporal dependencies, combined with higher-order spatial representations to model multi-modal future paths. While their study primarily addresses large-scale motion trajectories in urban and vehicular contexts, the modular architecture—consisting of a sequence encoder, latent generative representation, and autoregressive decoder—offers a flexible foundation for transfer to other motion domains. Building on this approach, the present work extends TrajLearn to the domain of house fly flight, which features rapid accelerations, abrupt directional changes, and small-scale 3D motion. By adapting the input embeddings to three-dimensional coordinates and fine-tuning the generative transformer on high-frame-rate insect flight sequences, this study demonstrates that transformer-based trajectory models can be effectively repurposed for predicting highly irregular, small-object motion that remains largely unexplored in the current literature [7].

3.9.2 Model-Based Tracking of Fruit Flies in Free Flight

Prior research has explored computer-vision based tracking of free-flying insects to extract detailed motion and pose information from high-speed video data. For example, Ben-Dov and Beatus developed a model-based tracking method for fruit flies that uses multi-view high-speed camera footage and a parameterized 3D fly model to estimate body and wing pose without strong kinematic assumptions. This approach demonstrated the feasibility of capturing complex insect motion data from high-speed imagery, which highlights both the potential and the challenges of vision-based tracking for small, fast moving targets like flies [5].

3.9.3 Laser-Based Insect Monitoring and Selective Targeting Concepts

In parallel, prior efforts have explored optical tracking and identification of insects in flight as a basis for selective intervention. Photonic Sentry describes a “Photonic Fence” concept that combines sensing with identification of insects and selective “shootdown” using lasers, framing the approach as an alternative to broad insecticide usage and highlighting applications in agriculture and vector-control contexts. The concept emphasizes discrimination between insect types and continuous monitoring as core capabilities, positioning the system as both a research tool (monitoring device) and a selective control mechanism. While the design choices and implementation details differ across systems and are outside the scope of this Phase-A report, such work is important as related context because it demonstrates real-world interest in combining real-time insect tracking, classification, and targeted response within a defined protected area [9].

3.9.4 Embedded Vision-Based Laser Neutralization Prototypes

Several experimental prototypes have explored the use of low-cost embedded platforms, such as Raspberry Pi systems, for vision-based mosquito detection and laser neutralization. These systems typically combine basic image processing with real-time actuation to identify and respond to flying insects in controlled environments. While such approaches demonstrate the feasibility of low-cost, autonomous insect control, they are generally limited by computational constraints, sensor latency, and reactive control strategies. As a result, their effectiveness is reduced when targeting fast and highly irregular motion, highlighting the need for predictive and learning-based methods as proposed in this work [11].

3.9.5 Trajectory Prediction with Noise-Resilient Transformer Models

In addition to object motion prediction within vision frameworks, recent work has explored robust trajectory forecasting in other dynamic domains. For example, Li *et al.* proposed a noise-robust autoregressive transformer with hybrid positional encoding for aircraft trajectory prediction, demonstrating improved long-term forecasting accuracy under real-world noise and complex temporal variations [6]. Although this work focuses on large-scale aircraft motion rather than small-scale insect flight, the methodological emphasis on handling noisy sequential data and enhancing transformer-based prediction robustness informs design considerations for highly irregular motion regimes such as those exhibited by house flies. By integrating noise regularization and autoregressive sequence modeling, such approaches illustrate the potential benefits of adapting advanced transformer architectures to motion prediction tasks beyond traditional domains, further motivating the present work's use of transfer-learned vision–temporal models and their adaptation to erratic, high-frame-rate insect motion.

3.9.6 Environmental and Biological Influences on Fly Motion

While many tracking and prediction approaches focus on visual and kinematic modeling, insect flight behavior is also strongly influenced by environmental and biological factors. Zahn and Gerry conducted an extensive field study on the diurnal flight activity of house flies (*Musca domestica*), demonstrating that flight intensity and temporal patterns depend on temperature, light intensity, time of day, and sex-specific behavioral responses [13]. Their results showed that fly activity increases rapidly with rising morning temperature and illumination, reaches a peak during mid-morning hours, and declines later in the day, with distinct activity profiles for male and female flies under varying environmental conditions.

These findings highlight that fly motion cannot be assumed to be stationary or purely random, but instead reflects a complex interaction between environmental stimuli and intrinsic biological traits. For vision-based tracking and prediction systems, this implies that motion statistics may shift over time even within the same scene, posing challenges for models that rely on fixed motion assumptions. This motivates the use of learning-based predictive models that can adapt to stochastic, context-dependent motion patterns, as pursued in the present work.

4 Project Description

The system employs an area-scan camera (0.3 MP, 815 FPS) equipped with a 75 mm fixed-focal industrial lens to capture high-speed video of house flies from a fixed vantage point. Video frames are streamed continuously to a processing unit, where a deep learning–based detection and tracking model identifies the precise spatial location of each fly in real time. The visual backbone of the model is initialized using transfer learning from a pretrained trajectory prediction framework [7], originally designed for generative modeling of motion trajectories using a transformer-based architecture, and is subsequently fine-tuned on high-frame-rate insect flight data. For each frame, the system outputs bounding boxes and centroid coordinates for all detected flies.
Detected positions are temporally aggregated across consecutive frames to form short-term fly trajectories, which are then processed by a transformer-based motion prediction module. This module, adapted from the pretrained TrajLearn architecture and retrained to model the highly non-linear dynamics of fly motion, predicts future fly positions over a short horizon. By learning temporal correlations and filtering noisy observations, the model enables accurate anticipation of rapid acceleration and abrupt direction changes that characterize fly flight behavior.
The predicted future positions are converted into control signals for a servo-actuated laser system, which is aligned to the anticipated target location to enable precise and autonomous neutralization. The full pipeline—from high-speed video acquisition to detection, trajectory construction, motion prediction, and laser actuation—is implemented in Python. PyTorch is used for all learning-based components, while custom hardware interfaces handle real-time control of the servo motors and laser. System timing is carefully synchronized to minimize end-to-end latency, enabling continuous autonomous operation. By integrating high-speed vision with transfer-learned predictive modeling based on TrajLearn, the system achieves reliable tracking and accurate targeting using predicted rather than purely reactive fly motion.

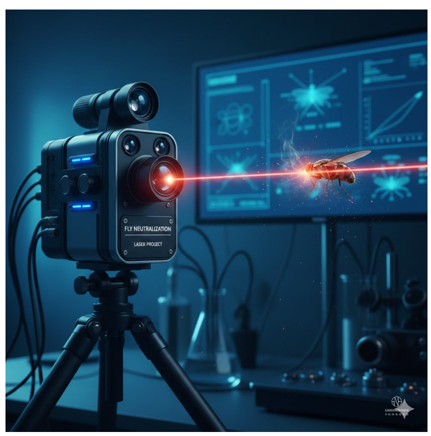

Figure 3: Illustration of a laser directed at a flying insect. This demonstrates the concept of selective neutralization of fast-moving flies using real-time vision-guided targeting.

4.1 Project Process Flow

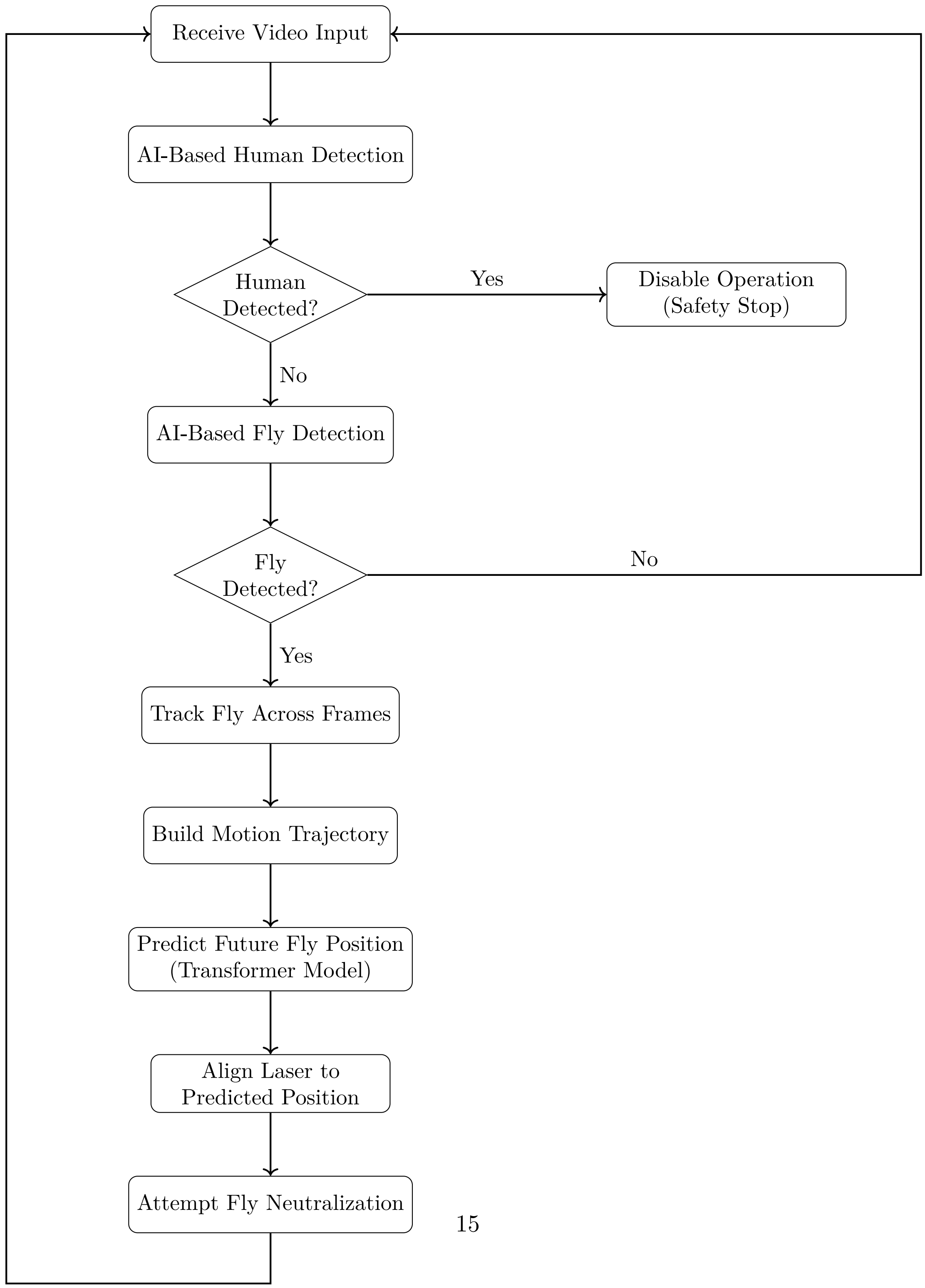

Figure 4: Flowchart of the autonomous predictive tracking and laser neutralization process with integrated safety control

4.2 System Architecture

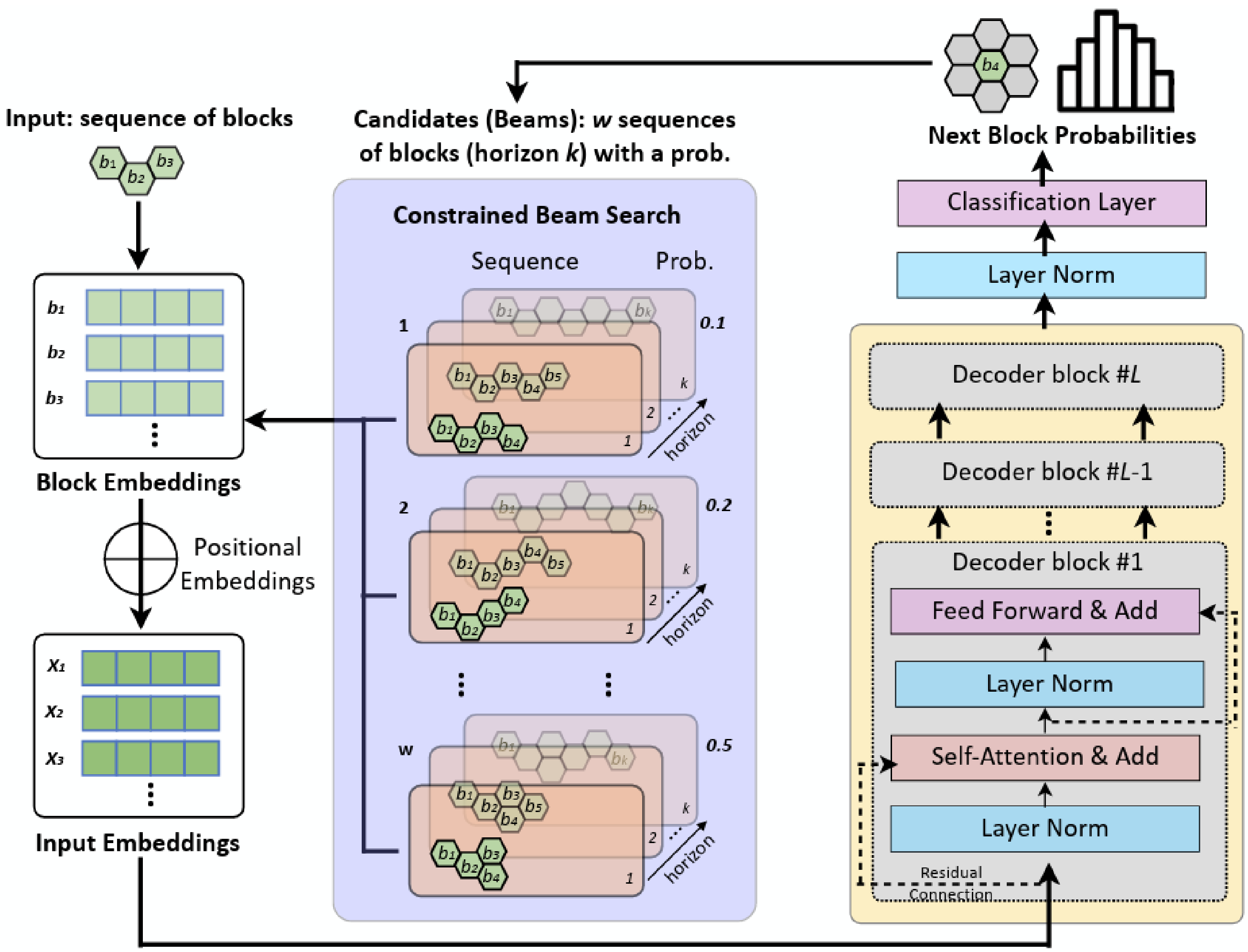

Figure 5: The TrajLearn architecture for trajectory prediction processes sequential motion data by first encoding past positions of agents into a latent representation using a Transformer Encoder Module, which captures temporal dependencies across observed time steps. A Latent Generative Module models the uncertainty and multi-modal nature of future trajectories, producing a distribution over possible future paths. The Transformer Decoder Module then autoregressively predicts the sequence of future positions for each agent, incorporating previous predicted positions at each decoding step to refine subsequent predictions. This design enables accurate forecasting of complex, non-linear motion. Additionally, TrajLearn supports transfer learning: pretrained encoder-decoder weights can be fine-tuned on new trajectory datasets, such as high-frame-rate insect flight data, allowing the model to adapt to domain-specific motion patterns while maintaining the capacity to generate plausible multi-step future trajectories.

4.3 Core Requirements

4.3.1 Functional Requirements

1. The system should be able to process video input from a camera or recorded video files.
2. The system should accurately detect flies in each video frame using AI-based computer vision methods.
3. The system should track detected flies across consecutive frames to generate motion trajectories.
4. The system should apply appropriate pre-processing techniques to enhance image quality and reduce noise.
5. The system should predict future fly positions based on recent motion history using a Transformer-based temporal prediction model.
6. The system should compensate for system latency using predicted future positions.
7. The system should direct a laser-based neutralization mechanism toward the predicted fly location.
8. The system should continuously monitor the video stream for human presence.
9. The system should disable all autonomous operation when human presence is detected.
10. The system should operate autonomously under controlled conditions.
11. The system should restrict the laser's motion and power to predefined safe limits to prevent damage to equipment or harm to humans.

4.3.2 Non-Functional Requirements

1. Autonomous operation without the need for continuous user interaction.
2. Accurate prediction of fly motion within an acceptable spatial error margin.
3. Consistent and reliable performance under varying environmental conditions, such as different lighting conditions and background complexity.
4. Efficient processing of video frames to support continuous operation.
5. Minimal end-to-end latency between visual input and predicted target position.
6. Robust handling of rapid direction changes and highly irregular fly motion.
7. Compatibility with different camera devices and video resolutions used for experimentation.
8. Stable frame rate suitable for tracking fast-moving targets (e.g., at least 20 frames per second).
9. Safety-first behavior, where detection of human presence immediately disables all autonomous operation.

4.4 Research Process

The research process included:

1. analysis of the limitations of reactive tracking approaches for fly motion;
2. investigation of learning-based vision models for small-object and human detection;
3. evaluation of temporal prediction methods for highly irregular motion;
4. formulation of a predictive tracking framework based on Transformer architectures;
5. definition of success metrics and safety constraints.

4.5 Algorithms and Methods

The research focuses on:

- neural-network-based fly detection models, using transfer learning from pretrained object detection networks to accelerate convergence on high-frame-rate insect data;
- neural-network-based human detection models for safety enforcement, leveraging pretrained human detection networks for robust performance;
- trajectory construction via temporal association of detected positions across frames;
- transformer-based temporal motion prediction, adapted from pretrained trajectory prediction models to capture the highly non-linear and rapid dynamics of fly flight.

4.6 Tools and Technologies

The system will be implemented using the following tools and technologies:

- **Programming language:** Python, for rapid prototyping and integration of AI models and hardware interfaces.
- **Machine learning frameworks:** PyTorch and TensorFlow, for developing and fine-tuning deep learning models for fly and human detection, as well as temporal motion prediction.
- **Computer vision libraries:** OpenCV, for video acquisition, pre-processing, and real-time image handling.
- **Data analysis and visualization:** NumPy and Matplotlib, for numerical computations, trajectory processing, and performance evaluation.
- **Hardware interfaces:** Custom Python scripts and libraries for real-time control of the servo-actuated laser system.

4.7 Success Metrics

Project success is evaluated using both quantitative evaluation metrics and qualitative system-level criteria.

4.7.1 Quantitative Metrics

- **Mean Absolute Error (MAE):**

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^{N} \left\| \mathbf{p}_{\text{pred}}(i) - \mathbf{p}_{\text{true}}(i) \right\|$$

- **Mean Squared Error (MSE):**

$$\text{MSE} = \frac{1}{N} \sum_{i=1}^{N} \left\| \mathbf{p}_{\text{pred}}(i) - \mathbf{p}_{\text{true}}(i) \right\|^2$$

- **Root Mean Squared Error (RMSE):**

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^{N} \left\| \mathbf{p}_{\text{pred}}(i) - \mathbf{p}_{\text{true}}(i) \right\|^2}$$

- **Cross-Entropy Loss (Training Objective):**

$$\mathcal{L} = -\frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} \log p_\theta(y_t \mid x_{\leq t})$$

Where:

- $\mathbf{p}_{\text{pred}}(i)$ and $\mathbf{p}_{\text{true}}(i)$ denote the predicted and ground-truth 3D fly positions at time step i,
- N is the total number of evaluated predictions
- y_t represents the discretized ground-truth trajectory token at timestep t
- $p_\theta(y_t \mid x_{\leq t})$ is the model-predicted probability for the next trajectory token
- $\mathcal{T}$ includes only valid (non-padded) timesteps used during training.

4.7.2 System-Level Success Criteria

- Accuracy of predicted future fly positions, quantified by the distance between predicted and observed locations.
- Demonstrated improvement of prediction-based targeting compared to reactive (non-predictive) approaches.
- Robust performance under noisy measurements and highly irregular fly motion.
- Successful alignment of the laser-based neutralization mechanism with the predicted fly position, enabling effective fly neutralization in controlled environments.
- Correct and consistent activation of safety constraints, including immediate inhibition of operation when human presence is detected or when prediction confidence is insufficient.

4.8 Testing and Evaluation Process

Evaluation is performed through a combination of analytical and experimental testing methods:

- Module-level testing of fly detection, tracking, human detection, and motion prediction components.
- End-to-end evaluation of the predictive pipeline, from visual input to laser targeting decision based on predicted fly position.
- Scenario-based testing that includes rapid and irregular fly motion, abrupt direction changes, occlusions, and safety-triggering cases involving human presence.
- Evaluation of laser targeting alignment accuracy by measuring the spatial error between the predicted fly position and the laser target location.
- Quantitative assessment of successful fly neutralization attempts in controlled scenarios, based on alignment accuracy and prediction timing.
- Comparative analysis against predefined success metrics, including prediction accuracy, latency feasibility, neutralization effectiveness, and safety constraint activation.

5 Expected Achievements

This chapter outlines the expected outcomes of the project, defines clear criteria for success, and discusses the primary engineering and research challenges anticipated during development. The objectives focus on predictive accuracy, robustness, safety, and real-time operation within controlled environments.

5.1 Outcome

The primary expected outcome of this project is the conceptual design and validation of an autonomous predictive tracking system capable of handling the highly irregular and fast motion of flies. The system is expected to demonstrate that learning-based temporal models, specifically Transformer-based architectures, can effectively predict near-future fly positions and compensate for system latency.
In addition, the project aims to present a safety-aware system architecture in which human presence detection reliably overrides autonomous operation. The final outcome includes a clearly defined software architecture, algorithmic justification, and performance analysis that together demonstrate the feasibility and advantages of predictive targeting over reactive approaches.

5.2 Criteria for Success

The project's success is evaluated using quantitative and qualitative criteria related to prediction accuracy, robustness, autonomy, and real-time performance.

5.2.1 Accurate Motion Prediction

The system should predict the future position of a fly with high accuracy, outperforming reactive tracking baselines. Prediction error is measured as the spatial distance between predicted and actual positions at the corresponding future time step. Consistent error reduction compared to non-predictive approaches indicates successful modeling.

5.2.2 Robustness and Generalizability

The system should maintain reliable performance under varying visual conditions and fly behaviors. Success is demonstrated if prediction error remains within acceptable bounds across different lighting, background complexity, and trajectory patterns, without manual adjustment.

5.2.3 Autonomous System Operation

The system must operate fully autonomously, with detection, prediction, and laser actuation running continuously in real time. All modules should be synchronized to ensure precise and reliable performance without user intervention.

5.2.4 Adaptability to Diverse Scenarios

The system should handle rapid accelerations, abrupt direction changes, and short stationary periods. Successful operation across these diverse motion patterns demonstrates the adaptability of the learning-based prediction approach.

5.2.5 Real-Time Processing

The complete pipeline—from video acquisition to prediction output—must meet timing constraints for fast-moving targets. Real-time capability is measured by end-to-end latency and the ability to generate predictions at a frame rate sufficient for timely actuation.

5.2.6 Noise and Uncertainty Tolerance

The system should remain stable under visual measurement noise due to lighting variations, sensor limitations, and background clutter. Success is defined as maintaining prediction accuracy above a defined threshold under such uncertainties.

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